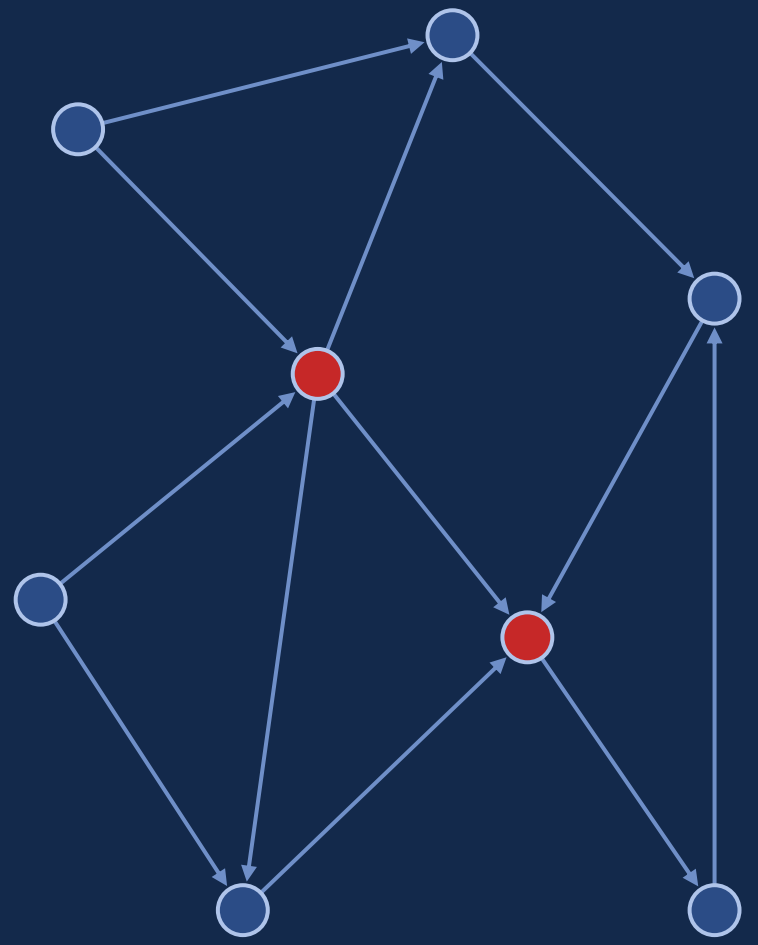
Towards Expressive Directed Graph Representations

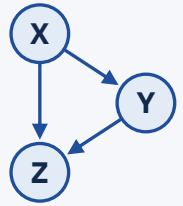
Magnetic Laplacians Meet Directed Walks



Yinan Huang
Georgia Institute of Technology
Joint work with Haoyu Wang, David F. Gleich and Pan Li

PART I · MOTIVATION

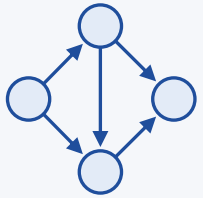
Networks are everywhere, and many are directed



Gene regulation

X regulates gene Y

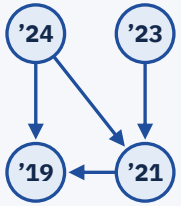
CAUSALITY



Protein interactions

A kinase activates its substrate

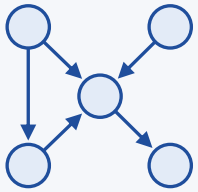
SIGNALING



Citation networks

New papers cite old ones

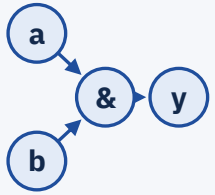
TIME



Social & trust

u follows or trusts v

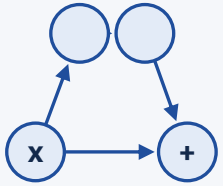
INFLUENCE



Circuits & programs

Signals flow from inputs to outputs

DEPENDENCY



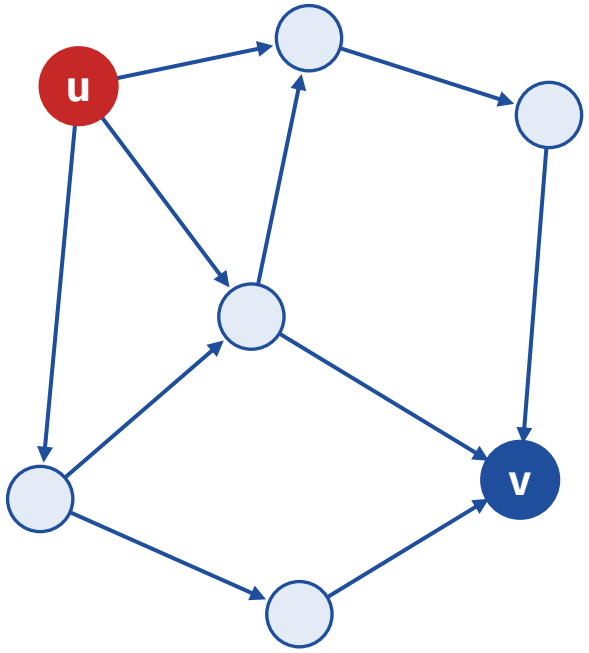
Computation graphs

Each step feeds the next

EXECUTION ORDER

Direction carries meaning; ignoring it throws that meaning away.

Why we need representations of graph structure



NODE role of u
(in/out degree, ...)

z_u



PAIR relation of u, v
(distances, walks, ...)

z_{uv}



GRAPH global structure

z_G



NETWORK ANALYSIS

- Motifs and communities
- Centrality and proximity
- Link prediction

MACHINE LEARNING

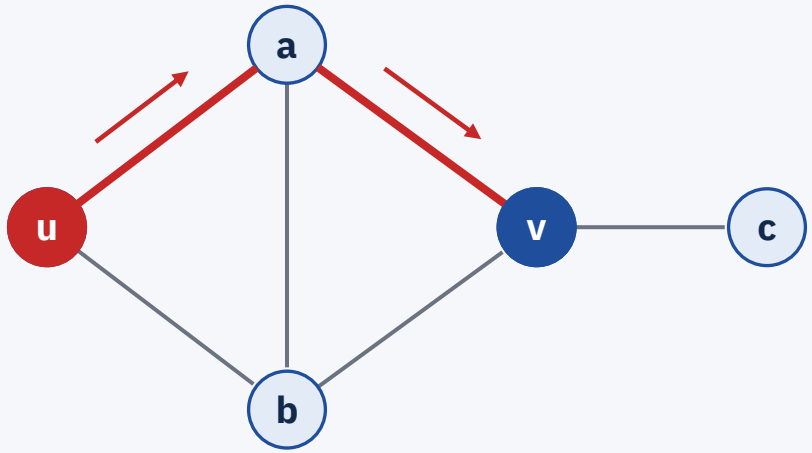
- Features for GNNs
- Positional encodings for graph transformers

Common need: structure → numbers, at node, pair and graph level.

Toolkit I: matrix powers count walks

Adjacency: $A_{uv} = 1$ if edge u–v, else 0

$[A^m]_{uv}$ = # walks of length m



$[A^2]_{uv} = 2$: via a or via b

1. **Distance**
   $\min \{ m : [A^m]_{uv} > 0 \}$
2. **Common neighbors**
   $[A^2]_{uv}$
3. **Triangles**
   $[A^3]_{uu} = 2 \times$ # triangles at u
4. **Power series in A**
   Katz, PageRank, heat kernel

Local and combinatorial: powers count small substructures.

Katz, Psychometrika 1953 · Page et al., 1999 · Chung, Spectral Graph Theory, 1997

Toolkit II: the spectrum, and how it links to walks

GRAPH SPECTRUM

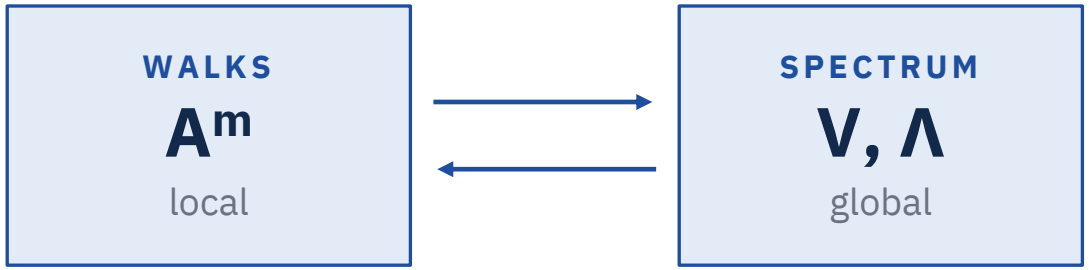
$$A = V \Lambda V^T$$

real eigenvalues, orthonormal eigenvectors

Normalized Laplacian $L = I - D^{-½}AD^{-½}$

WHAT IT TELLS US

- # zero eigenvalues = # components
- Fiedler vector (2nd eigenvector): its signs split two clusters
- Eigenvectors → embeddings, PEs



$$A^m = V\Lambda^m V^T$$

more generally, $f(A) = V\, f(\Lambda)\, V^T$

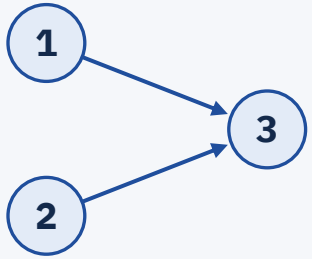
Eigenpairs encode every walk count.

Template: walks ⇄ spectrum, two views of one object.

Fiedler, 1973 · Chung, Spectral Graph Theory, 1997 · Belkin & Niyogi, 2003

Why this breaks for directed graphs

WALK SIDE

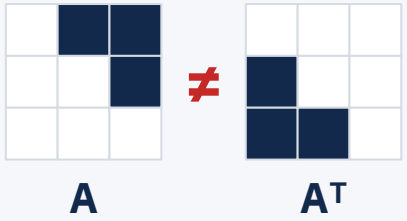


1 and 2 share a successor, yet

$[A^m]_{12} = [A^m]_{21} = 0$

- A^m sees only forward walks
- Mixing A and A^T: 2^m words of length m
- $(A + A^T)^m$ sums them: direction lost

SPECTRAL SIDE



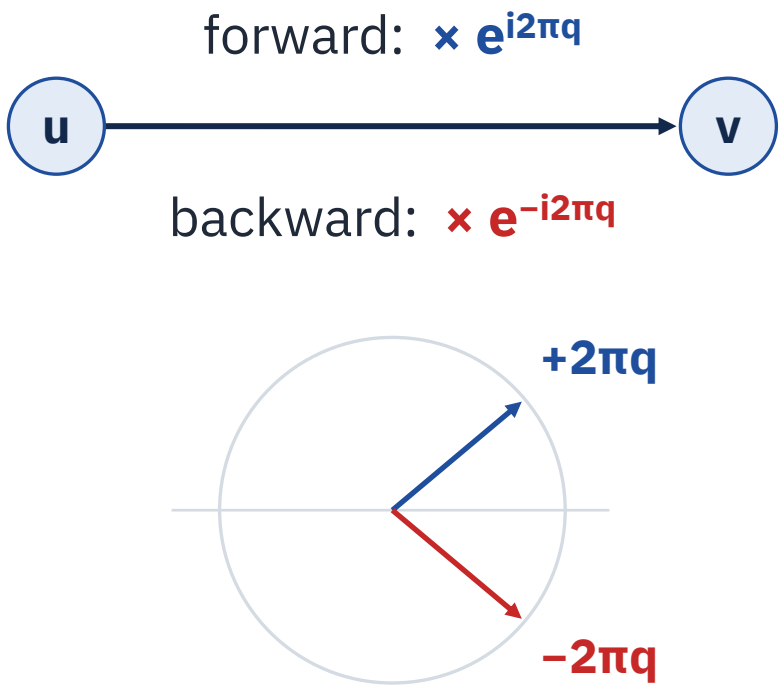
No orthogonal eigendecomposition

$A = V\Lambda V^T$

- Eigenvalues can be complex
- Eigenvectors need not be orthogonal, or even complete
- Random walks are not reversible

The link $A^m = V\Lambda^m V^T$ breaks. Nothing connects the two views.

The magnetic matrix: direction as a phase



$$A_q = A\, e^{i2\pi q} + A^T e^{-i2\pi q}$$

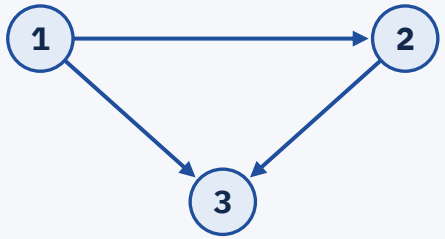
THE MAGNETIC POTENTIAL q

One knob: how much phase each directed edge carries

- **q = 0**: the undirected graph $A + A^T$
- **q = ¼**: purely antisymmetric, $i(A - A^T)$
- **Magnetic Laplacian:** $L_q = I - D^{-½} A_q D^{-½}$

Direction becomes a phase, and the potential q sets its size.

Why the magnetic matrix matters



$$A_q = \begin{pmatrix} 0 & e^{i\theta} & e^{i\theta} \\ e^{-i\theta} & 0 & e^{i\theta} \\ e^{-i\theta} & e^{-i\theta} & 0 \end{pmatrix}$$

$\vartheta = 2\pi q$ · mirror entries are conjugates

✓ Keeps direction
u → v and v → u get conjugate phases

✓ Real eigenvalues
Hermitian: equal to its conjugate transpose

✓ Full eigenbasis
Orthonormal, for every graph

A WIDELY USED TOOL

Physics
Electrons in a magnetic field
Harper 1955 · Shubin 1994

Community detection
Directed networks
Fanuel et al., PRE 2017

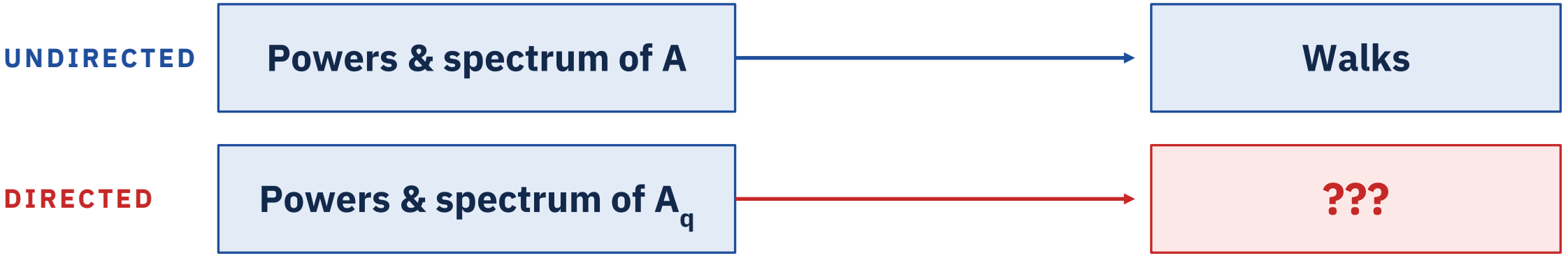
Network analysis
Visualization, network comparison
Fanuel et al. 2018 · de Resende & Costa 2020

Graph learning
MagNet, directed graph transformers
Zhang et al. 2021 · Geisler et al. 2023

Used mostly via eigenvectors. What local directed structure does it encode?

PART I · THE PROBLEM

The question of this talk



What directed substructures do magnetic matrices encode?

Q1

What do powers of A_q count?

Analog of walk counts

Q2

How much information do they carry?

How many potentials q?

Q3

How can we use them?

Network analysis and learning

Roadmap

I Motivation: magnetic matrices and directed substructures

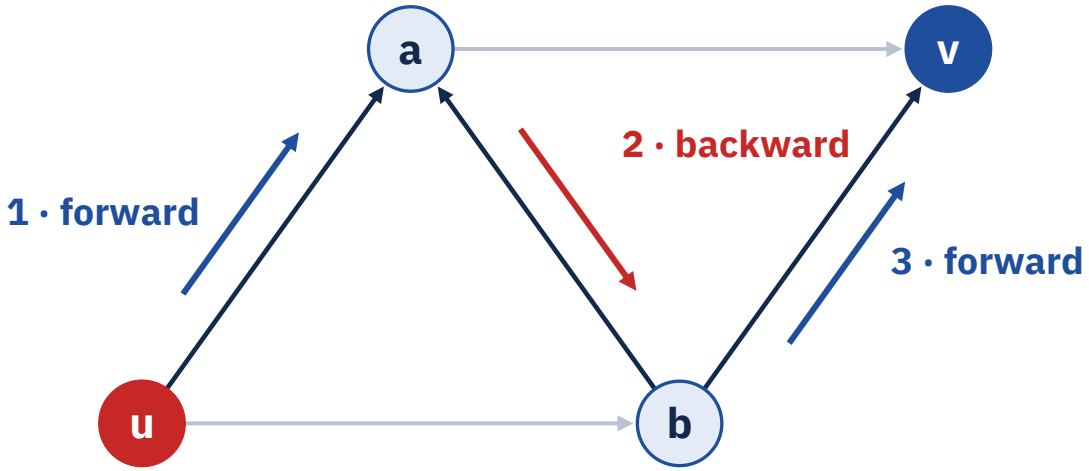
II **Walk profiles and a Fourier duality** — Q1

III Compression, directed cycles, link prediction — Q2 · Q3

IV Positional encodings for directed GNNs and transformers — Q3

II–III: Huang, Gleich, Li · Powers of Magnetic Graph Matrix: Fourier Spectrum, Walk Compression, and Applications · PNAS 2026
IV: Huang, Wang, Li · What Are Good Positional Encodings for Directed Graphs? · ICLR 2025

Bidirectional walks: step with or against an edge



u → a → b → v: **m = 3, k = 2 forward**

BIDIRECTIONAL WALK

Each step uses an edge either forward (along its direction) or backward (against it).

BACK TO PART I



Now 1 reaches 2: one forward step, then one backward step.

Backward steps allowed, and tracked.

Walk profiles: count walks by forward steps

$\Phi_{uv}(m, k)$ = # length-m walks u → v with k forward steps

k = 2

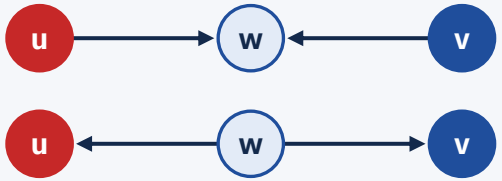
$\Phi(2, 2) = A^2$



u reaches v

k = 1

$\Phi(2, 1) = AA^T + A^TA$



common successor / predecessor

k = 0

$\Phi(2, 0) = (A^T)^2$



v reaches u

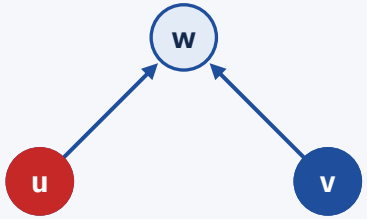
k = m: directed walks; **sum over k:** undirected walks

Walk profiles capture directed substructures



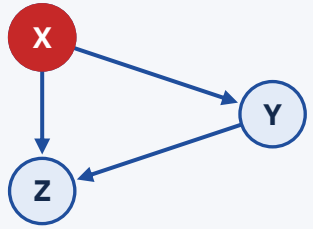
Directed distances

- Shortest path: smallest m with $\Phi_{uv}(m, m) > 0$
- On DAGs, longest path: the largest such m



Common successors and predecessors

- Counted by $\Phi_{uv}(2, 1)$
- Used in logic and data-flow analysis

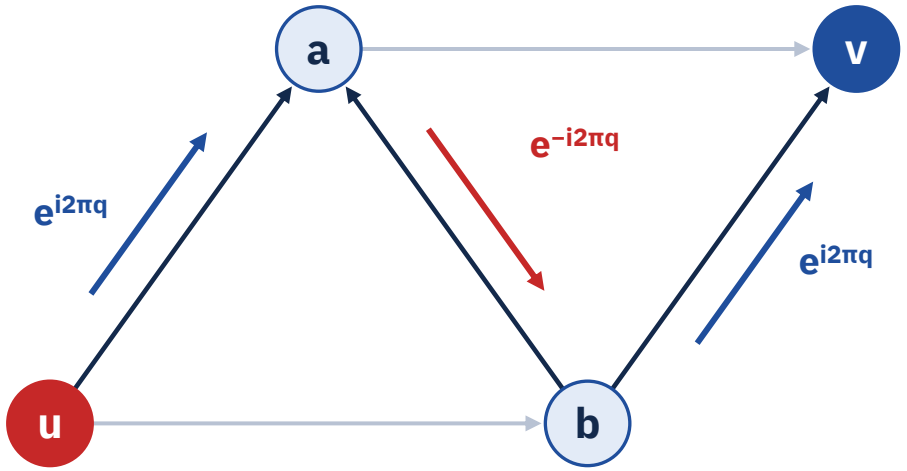


Feedforward loops

- Closed walk X → Y → Z ← X: in $\Phi_{XX}(3, 2)$
- Key motif in gene regulation

Walk profiles: the directed substructures we target.

Magnetic powers: phases multiply along a walk



$$e^{i2\pi q} \cdot e^{-i2\pi q} \cdot e^{i2\pi q} = e^{i2\pi q\,(2k-m)}$$

Depends only on k, the number of forward steps

EXPAND THE POWER

$$(Ae^{i2\pi q} + A^T e^{-i2\pi q})^m$$

- Sum over all 2^m words in A and A^T
- A word with k copies of A carries $e^{i2\pi q(2k-m)}$
- Group the words by k:

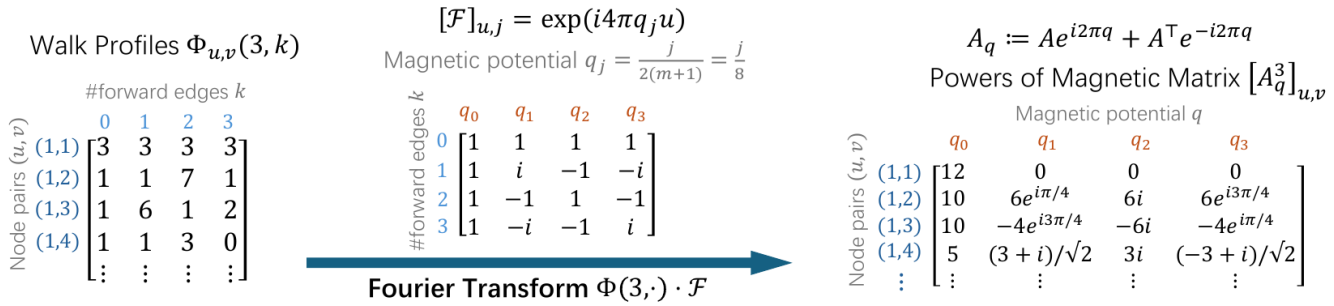
$$[A_q^m]_{uv} = \sum_{k=0}^{m} e^{i2\pi q\,(2k-m)}\ \Phi_{uv}(m,k)$$

Magnetic powers count bidirectional walks, weighted by a phase.

Magnetic powers are Fourier transforms of walk profiles

$$[A_q^m]_{uv} = e^{-i2\pi qm} \sum_{k=0}^{m} e^{i4\pi qk} \, \Phi_{uv}(m,k)$$

- **q is a frequency** over k, the number of forward steps
- **One potential = one Fourier coefficient**
- **q = 0** is the zero frequency: undirected walks $(A + A^T)^m$



Example from the paper: m = 3, potentials q = j/8

Answer to Q1: magnetic powers count directed walks, sorted by frequency.

One potential is not enough

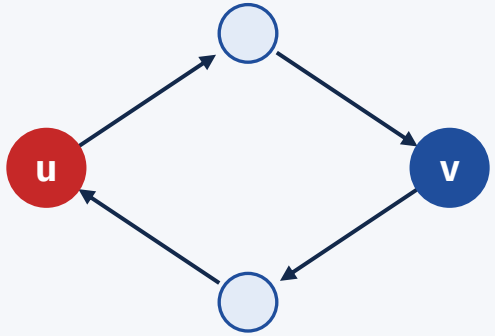
v cannot be reached



$\Phi_{uv}(2, \cdot) = \mathbf{(0, 0, 0)}$

$[A_q^2]_{uv} = \mathbf{0}$

Two paths, opposite directions



$\Phi_{uv}(2, \cdot) = \mathbf{(1, 0, 1)}$

$[A_q^2]_{uv} = \mathbf{i + (-i) = 0}$

Take q = 1/8, m = 2

- A single $[A_q^m]_{uv}$ squeezes m + 1 counts into one complex number
- Different walk profiles can give the same value
- Here the forward path (phase i) and the backward path (phase –i) cancel

One potential = one Fourier sample: it cannot recover the walk profile.

Several potentials recover the walk profile exactly

SAMPLE SEVERAL POTENTIALS

$A_{q_0}^{m}$, $A_{q_1}^{m}$, …, $A_{q_j}^{m}$

inverse DFT

FULL WALK PROFILE

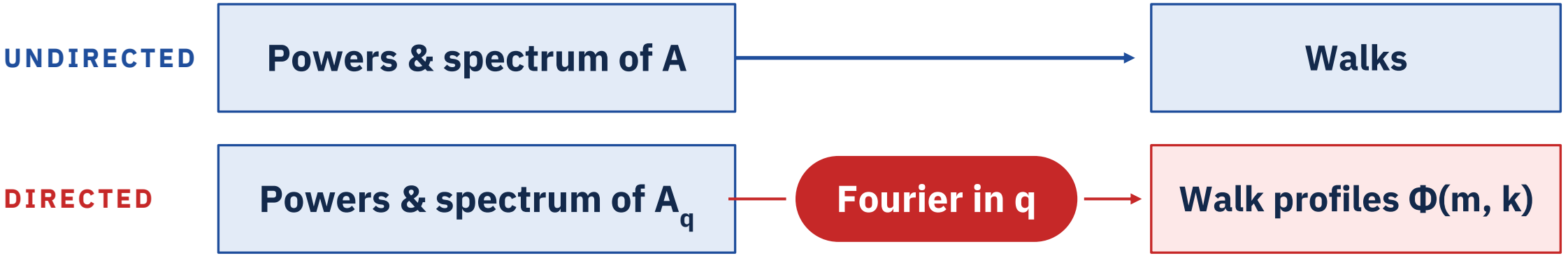
$\Phi(m, 0)$, $\Phi(m, 1)$, …, $\Phi(m, m)$

$$\Phi_{uv}(m,k) = \frac{1}{m+1}\sum_{j=0}^{m} e^{-i2\pi q_j(2k-m)}\,[A_{q_j}^{m}]_{uv}, \qquad q_j = \frac{j}{2(m+1)}$$

- Φ real ⇒ **[m/2] + 1 potentials in [0, ¼] suffice**
- Conversely, Φ gives A_q^m for every q: **same information**

Exact is costly: m = 50 needs 26 potentials. Real networks? → Part III

Answer to Q1: walks meet magnetic matrices



Spectrum: $A_q^m = V\Lambda^m V^*$, so eigenpairs at several q encode walk profiles too

q is a frequency

Each potential reads one Fourier coefficient of the walk profile

[m/2] + 1 potentials

Enough to recover the full walk profile exactly

Random-walk version

Walk profiles become random-walk probabilities

Roadmap

I Motivation: magnetic matrices and directed substructures

II Walk profiles and a Fourier duality — Q1

III Compression, directed cycles, link prediction — Q2 · Q3

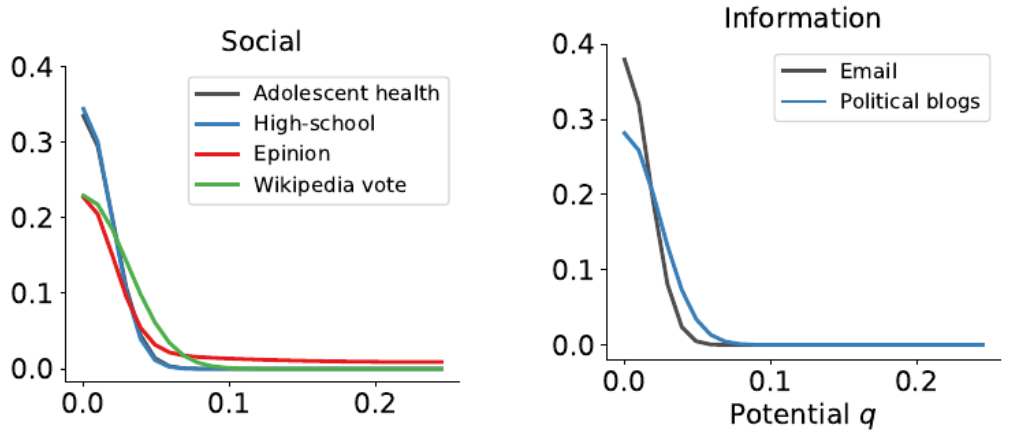
IV Positional encodings for directed GNNs and transformers — Q3

II–III: Huang, Gleich, Li · Powers of Magnetic Graph Matrix: Fourier Spectrum, Walk Compression, and Applications · PNAS 2026
IV: Huang, Wang, Li · What Are Good Positional Encodings for Directed Graphs? · ICLR 2025

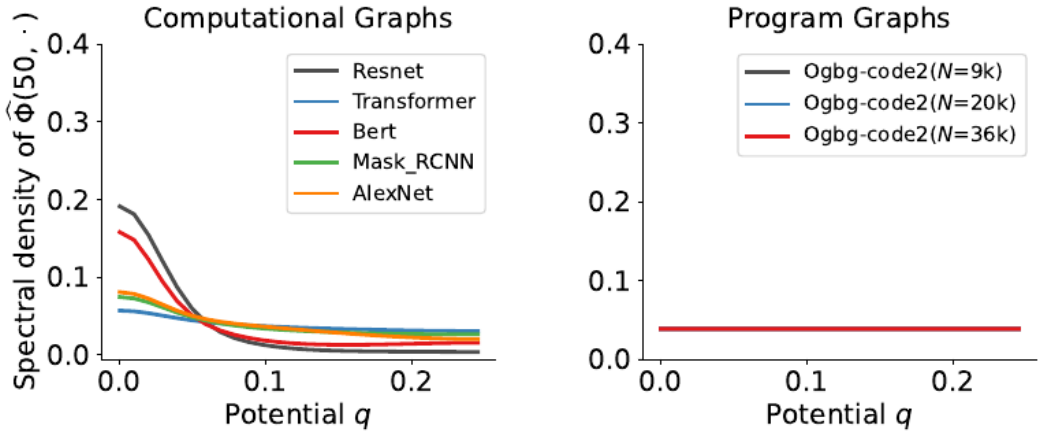
Where does the walk-profile energy sit?

Energy at q: share of $|[\hat{A}_q^m]_{uv}|^2$ at that q, averaged over pairs (m = 50)

Concentrated at low q



Spread out



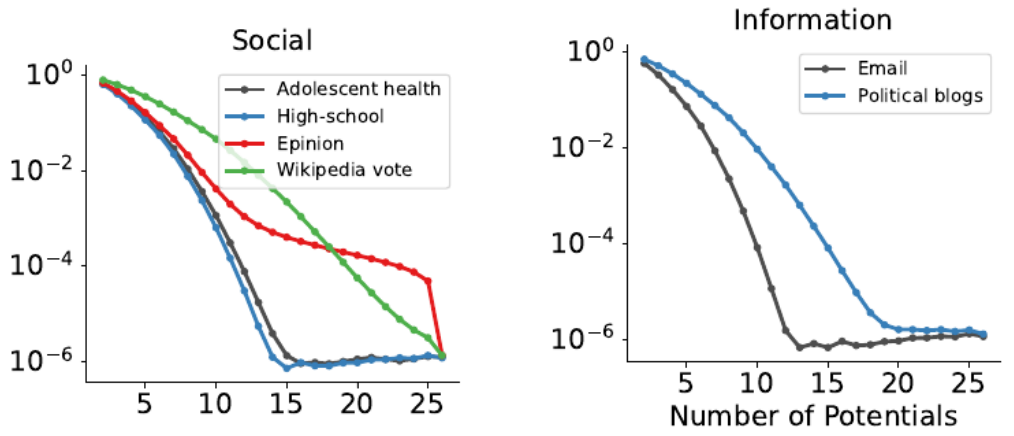
26 real networks, 8 domains, 10^2 to 10^6 nodes; four domains shown

In most networks, the energy sits in a few low potentials.

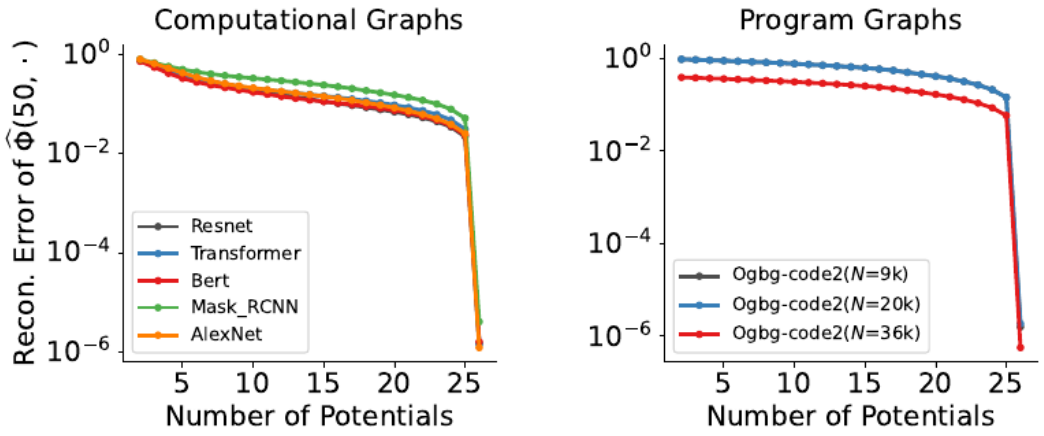
A few potentials reconstruct the walk profile

Error: relative gap to the walk profile rebuilt from the Q lowest potentials

Fewer than 10 potentials suffice



Barely compress



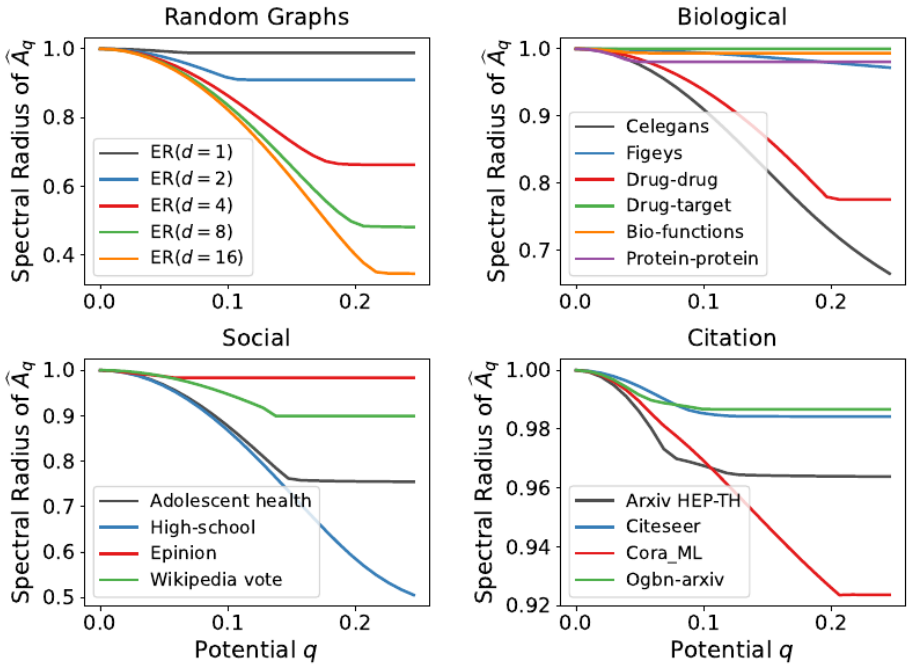
Exact recovery needs 26 potentials for m = 50

Answer to Q2: a few potentials carry most of the directed structure.

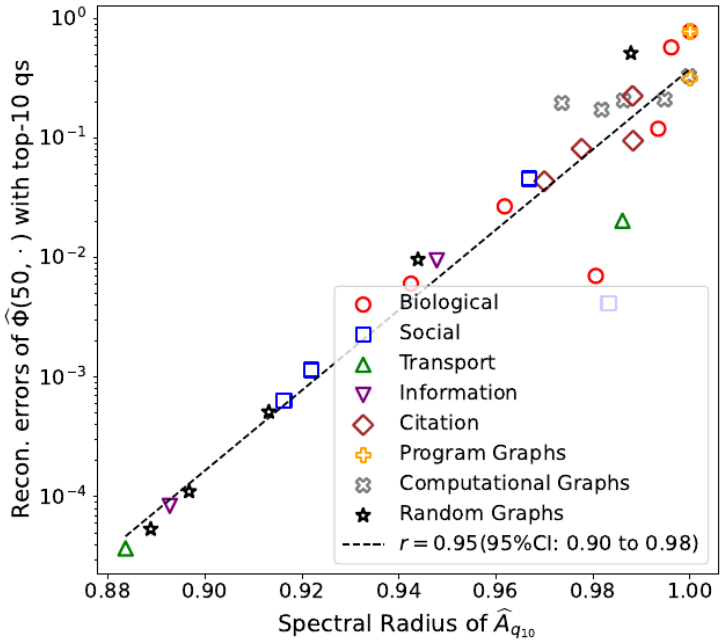
What predicts compression: the spectral radius

$\rho(\hat{A}_q)$: spectral radius of $D^{-1}A_q$; magnitudes shrink like $\rho(\hat{A}_q)^m$

ρ decays with q

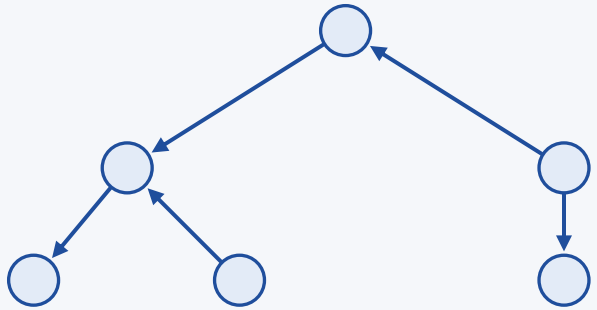


Faster decay → better compression (r = 0.95)



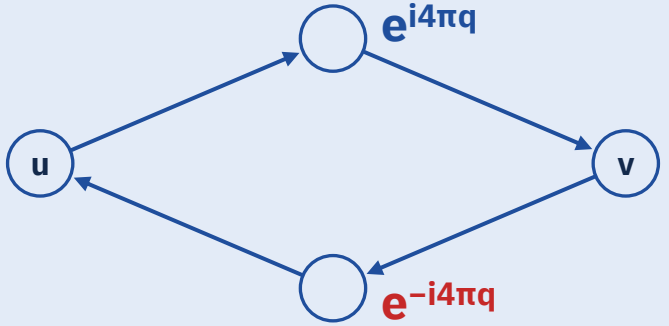
Intuition: cycles make phases interfere

TREE-LIKE: NO CYCLES



- Phases absorbed into the nodes
- $|[A_q^m]_{uv}|$ same for every q
- **Flat:** no compression

WITH CYCLES

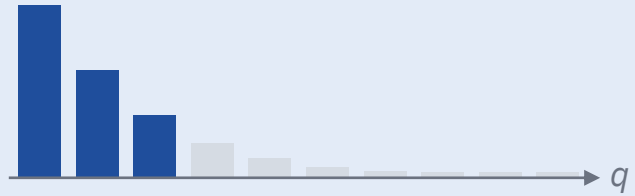


- Forward and backward walk: opposite phases
- Sum $2\cos(4\pi q)$: they cancel
- **Low-q energy:** compressible

Mismatched cycles ⇒ ρ drops faster ⇒ easier compression

Answer to Q2: a few potentials suffice

MOST REAL NETWORKS (WITH CYCLES)



Energy at low q: keep a few potentials

TREE-LIKE (E.G. PROGRAM GRAPHS)



Flat: every potential matters

Few potentials

m = 50: fewer than 10 give < 5% error, vs. 26 for exact recovery

Why: cycles

Phases of walks around cycles interfere, so high-q terms cancel

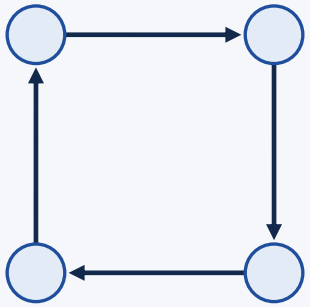
Predictable

Spectral radius $\rho(\hat{A}_q)$ tells how fast; r = 0.95 with the error

On most non-tree graphs, a few potentials represent walk profiles well.

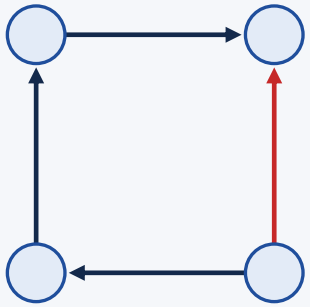
Which directed cycles can walks detect?

All forward



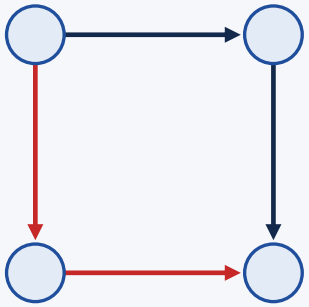
✓ detectable

One reversed



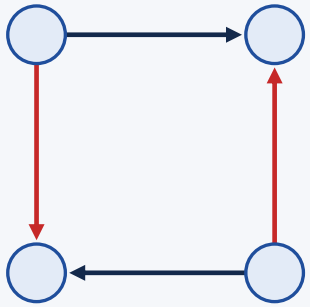
✓ detectable

Two reversed, adjacent



✗ not detectable

Two reversed, apart



✗ not detectable

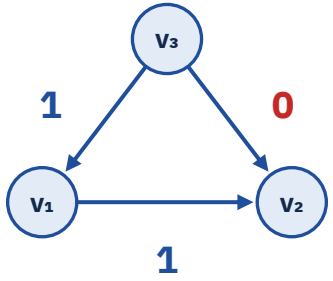
Theorem. ≤ 1 reversed edge: detectable; ≥ 2: cannot be located.

Feedforward loops have one reversed edge, so they are detectable.

Motif detection: a few potentials suffice

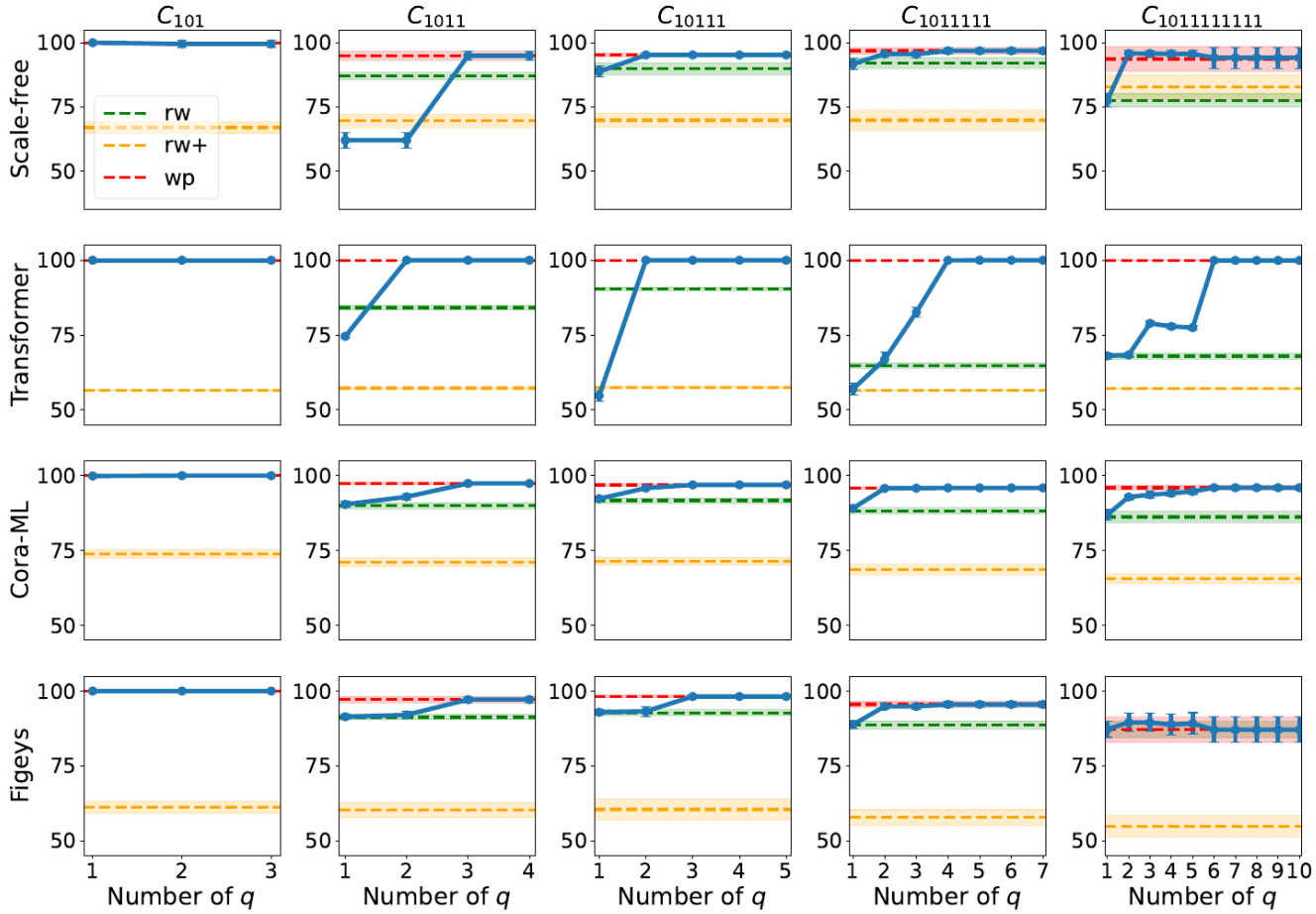
Cycle class $\mathbf{C_{b_1 \ldots b_m}}$

b_i = 1: forward edge, 0: reversed



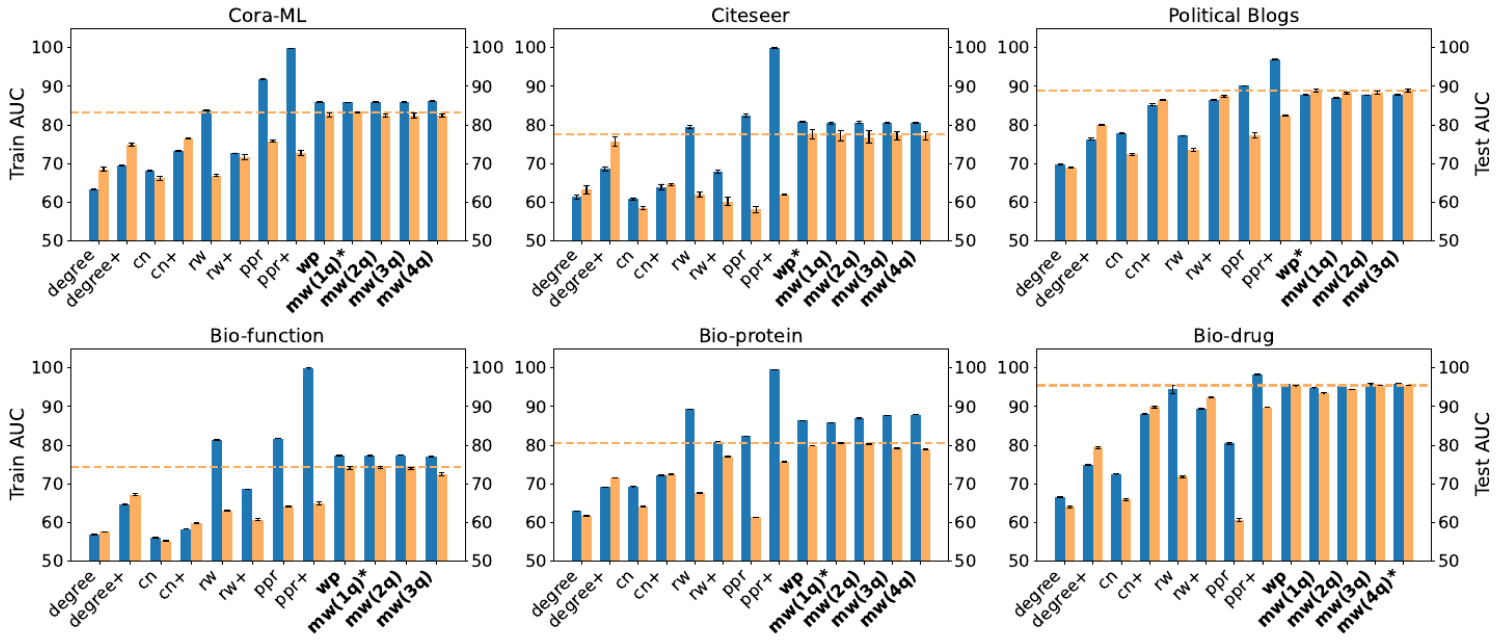
$\mathbf{C_{101}}$ = feedforward loop

- Task: is u on the cycle? (AUC)
- **Blue:** magnetic, more q →
- **Red:** walk profile
- **Green/orange:** baselines
- **A few q reach red**



Link prediction with magnetic features

- Features $[A_q^m]_{uv}$, $m \le 4$
- **One potential already matches walk profiles**
- Beats degree, common-neighbor and PageRank features



Compression in practice: even one potential captures useful directed structure.

Roadmap

I Motivation: magnetic matrices and directed substructures

II Walk profiles and a Fourier duality — Q1

III Compression, directed cycles, link prediction — Q2 · Q3

IV Positional encodings for directed GNNs and transformers — Q3

II–III: Huang, Gleich, Li · Powers of Magnetic Graph Matrix: Fourier Spectrum, Walk Compression, and Applications · PNAS 2026
IV: Huang, Wang, Li · What Are Good Positional Encodings for Directed Graphs? · ICLR 2025

Positional encodings: where is each token?

Transformer input = token + positional encoding (PE); attention ignores order.

SEQUENCE

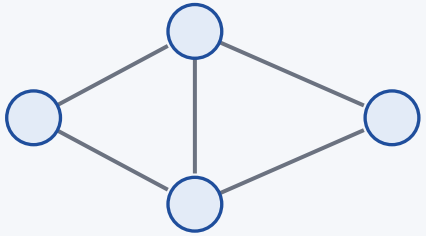


Total order

$$z_k = [\sin(\omega k), \cos(\omega k)]$$

positions are frequencies

UNDIRECTED GRAPH



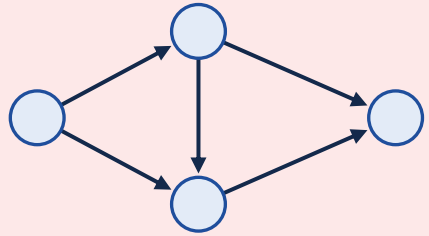
No order, only symmetry

$$L = V \Lambda V^T$$

$$z_u = [V]_{u,\,1:k}$$

first k eigenvectors

DIRECTED GRAPH



Partial order, direction

?

Question: what makes a positional encoding good for directed graphs?

Vaswani et al., NeurIPS 2017 · Dwivedi & Bresson, 2021 · Kreuzer et al., NeurIPS 2021

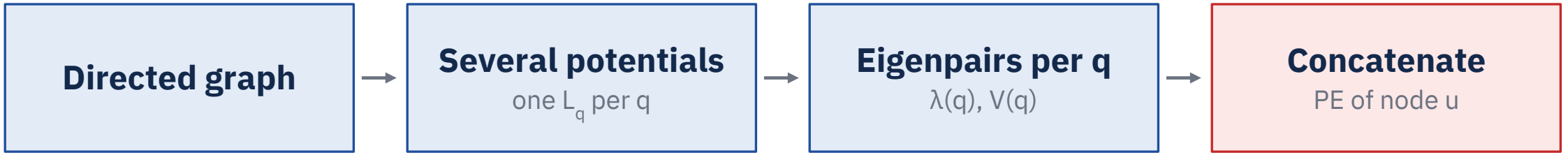
What makes a directed PE good?

Yardstick: walk profiles, the directed substructures from Part II.

Criterion: from the encodings of u and v, we can compute $\Phi_{uv}(m, k)$

Symmetrized Laplacian PE Dwivedi et al., JMLR 2023	✗	direction is lost
SVD PE Hussain et al., KDD 2022	✗	A^2 needs other nodes' vectors
Magnetic Laplacian PE, one q Geisler et al., ICML 2023	✗	one Fourier sample (Part II)
Multi-q magnetic PE this work, ICLR 2025	✓	recovers walk profiles

Multi-q magnetic Laplacian PE



Theorem. With [m/2] + 1 distinct potentials in [0, ¼), multi-q PE computes every walk profile of length ≤ m.

Why: eigenpairs give A_q^m at each q; Part II's inverse DFT does the rest.

- A simple change to magnetic PE, with a provable gain
- Cost: one eigendecomposition per q; 10 potentials ≈ 1.5–3× training time

Complex eigenvectors need stable processing

Problem: eigenvectors are not unique (rotations, phases) and unstable.

$$z_{\mathrm{PE}} = \rho\left(V_q \,\mathrm{diag}(\phi(\lambda_q))\, V_q^*\right)$$

- V_q eigenvectors of L_q
- V_q^* conjugate transpose
- φ equivariant network on eigenvalues
- ρ equivariant network → PE

Invariant
Same output for any rotation or phase of the eigenvectors

Stable
φ Lipschitz: small graph change, small PE change

Continuous in q
q = 0 recovers stable Laplacian PE

Results: expressing directed relations

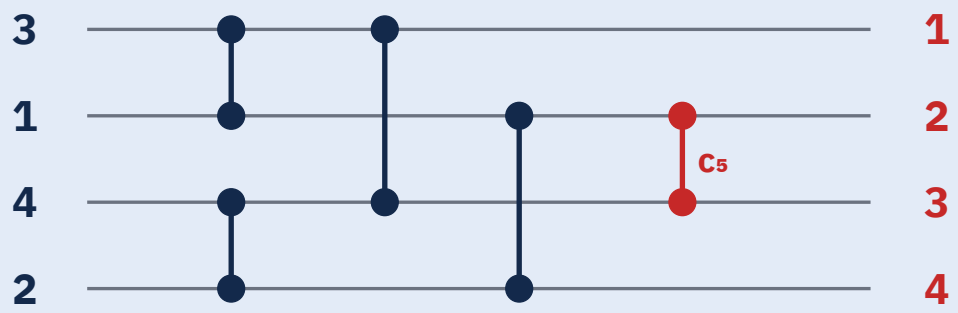
Task: random DAGs (train 16–63 nodes, test 64–71). From the PEs of u, v predict the walk profile (m = 4) and shortest-path distance. RMSE ↓

	Laplacian PE	Best single q	Multi-q
Walk profile	**0.326**	**0.040**	**0.002**
Shortest-path distance	**0.355**	**0.124**	**0.016**

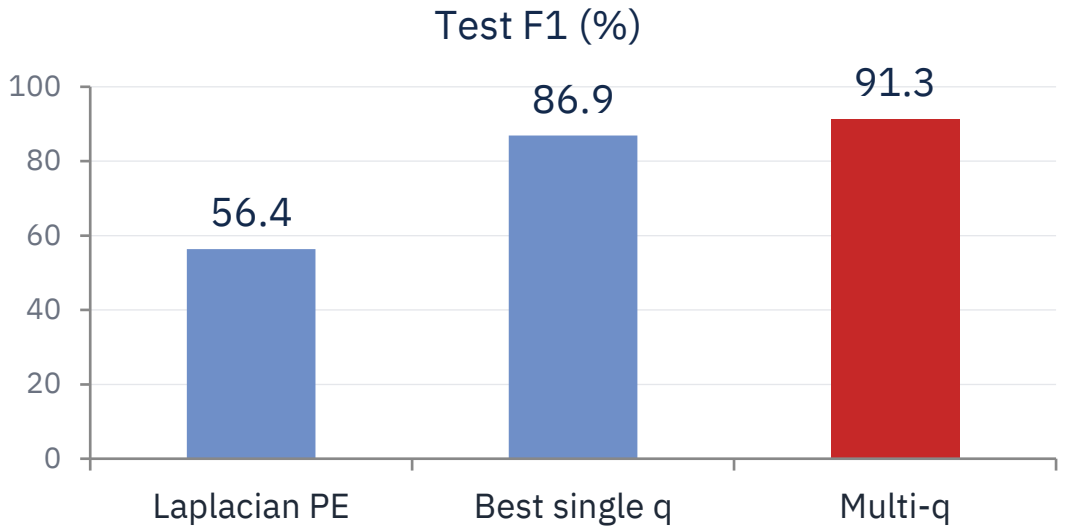
Baselines miss walk profiles, even shortest paths; multi-q gets them.

Results: sorting networks

TASK: IS THE SORTING NETWORK VALID?



Comparator: smaller value goes up
Valid = sorts every input
Drop c_5: 3,1,4,2 → 1,3,2,4 ✗
As a DAG: comparators = nodes



Longer at test: train 7–11 variables, test 13–16; multi-q holds, single q drops

Direction-aware positions help where the order of operations matters.

Results: real-world circuits

Analog circuits
10,000 op-amps: gain, bandwidth, phase margin (RMSE)

High-level synthesis
18,750 C/C++ programs: FPGA DSP, LUT use (MSE)

PE	Gain	BW	PM	DSP	LUT
No PE	0.386	4.594	1.123	2.232	1.939
Laplacian PE	0.391	4.153	1.135	2.267	1.786
Magnetic PE, best single q	0.383	4.113	**1.099**	2.256	1.768
Multi-q magnetic PE	**0.371**	**4.051**	1.116	**2.207**	**1.735**

Bidirected-GIN backbone, stable processing; lower is better; best in red

Multi-q is best on 4 of 5 targets; similar trends with other backbones.

SUMMARY

Magnetic Laplacians meet directed walks

Q1 · THEORY

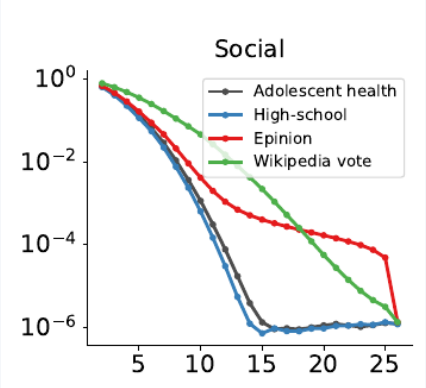
Magnetic powers = Fourier of walk profiles

$$[A_q^m]_{uv} = e^{-i2\pi qm} \sum_{k=0}^{m} e^{i4\pi qk} \Phi_{uv}(m,k)$$

q is a frequency over forward steps

Q2 · COMPRESSION

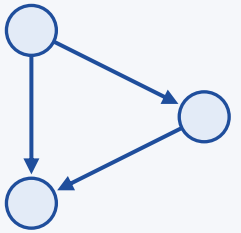
A few potentials suffice



Governed by the spectral radius

Q3 · APPLICATIONS

Cycles, links, expressive PEs



Feedforward loops ✓

Walk-profile error **0.040 → 0.002**

Sorting networks F1 **91.3**

Magnetic powers & spectra ⇄ Fourier in q ⇄ walk profiles

Thank you

Questions?

Powers of Magnetic Graph Matrix: Fourier Spectrum, Walk Compression, and Applications
Y. Huang, D. F. Gleich, P. Li · PNAS 2026 · github.com/Graph-COM/Walk_Profiles

What Are Good Positional Encodings for Directed Graphs?
Y. Huang, H. Wang, P. Li · ICLR 2025 · github.com/Graph-COM/Multi-q-Maglap

On the Stability of Expressive Positional Encodings for Graphs
Y. Huang, W. Lu, J. Robinson, Y. Yang, M. Zhang, S. Jegelka, P. Li · ICLR 2024